Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Full-Stack Product Engineer with 3 years of industry experience shipping internal products end to end in regulated financial services: **ReactJS** frontends, **Python FastAPI** backends, **PostgreSQL** and **AWS**, deployed through a one-click **Jenkins** and **GitLab CI/CD** pipeline. Built the first applications as the only engineer alongside my manager, working directly with Client Service, Compliance and Investment users to find workflow bottlenecks and ship the fix, such as a pipeline that parses **300+** daily rating-agency emails into structured data and saves **60** analyst hours a month. Later acted as de facto team lead for up to 4 interns. Takes requirements from ambiguous requests through to release and turns user feedback into the next iteration. Completed an MSc in Data Science (awaiting result); twice recognised with Best Team Project.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Full-Stack Product Development

- Specified, designed and built the company's first GenAI applications end to end, initially with only my manager: a **ReactJS** frontend served by **Nginx**, **Python FastAPI** REST API **microservices** (**uvicorn**) on a separate **AWS EC2** instance, and **RAG** (**LlamaIndex**, **Azure OpenAI**) over a separate **pgvector** database per department, for Compliance, Client Service and Investment, so each team retrieved only from its own documents.
- Released to real users and changed the products based on what they found, with Compliance reviewing every application; built **tool-calling** agents whose **Python** functions run **SQL** queries to compare funds and generate PDF reports on request, and **ExpressJS** APIs during development to serve the fund-comparison calculations because the Python APIs were slower, with the final backend in **FastAPI**.
- Recreated the **Power BI** dashboards in the **ReactJS** frontend because Power BI licensing was too costly to scale to the nationwide Client Service team, giving the whole organization access to the data at lower cost; the dashboards had first moved financial analysts from Excel to **Power BI**, improving efficiency by **10** hours a week.

Workflow Automation, Data & Databases

- Designed the **Normalized SQL Schema** (**PostgreSQL**) for the internal analytics platform to the analysts' requirements, with **Optimized SQL** queries and **SQL Functions** and **Procedures**; wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with the legacy analyst data.
- Removed a manual monitoring task for the analysts by building a **Python** pipeline, scheduled with **EventBridge**, **Lambda** and **AWS SageMaker**, that uses **Azure OpenAI** to parse over **300** daily emails from credit rating agencies into structured data (first version used **AWS Textract**, replaced when a rating agency began sending emails as images), fuzzy-matches company names against the ones analysts track and sends a daily summary, saving **60** analyst hours a month.
- Analysed customer data with **Python Pandas** and identified buying patterns with **Market Basket Analysis** (**Apriori**) for the marketing team.

Cloud Infrastructure, Security & Delivery

- Packaged the **FastAPI** microservices (uvicorn) as **Docker** containers on **AWS** (EC2, ECR), provisioned **EC2** instances using **CloudFormation** templates prebuilt by the security team, and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.
- Protected the internal applications with **RBAC** using a **Python** authentication function against the company **LDAP** server and **AWS Security Groups** limiting access to company private IPs; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**, and coordinated with them on which ports to whitelist.
- Worked directly with Client Service, Compliance and Investment to turn requests into clear requirements and feedback into releases; acted as de facto team lead for up to 4 interns in my final year; recognised with Best Team Project twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **Wingman:** AI dating-coach web app with a **ReactJS** (Vite, Tailwind) frontend and a Python **FastAPI** backend calling LLMs via NVIDIA NIM. Rate limits shaped the design: calls serialised behind a semaphore, 12 evaluations instead of 108, results streamed to the UI over server-sent events and cached by input hash; LLM output treated as untrusted and validated with Pydantic. GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. A tool built as FastAPI microservices with a React frontend that ranks cancer cell lines by target inclusion and exclusion genes and returns the evidence behind each pick. Personal focus: the data preprocessing pipeline that joins raw genomic sources into clean tables with integrity checks, the scoring algorithm and the RAG layer; refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/

ACHIEVEMENTS

- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** REST APIs, Git, Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker), Nginx, Linux, LDAP, Azure OpenAI, LlamaIndex, Claude Code, Power BI, VS Code.
- **Languages:** Python (FastAPI, uvicorn, Pydantic, pandas, scikit-learn), SQL, JavaScript/TypeScript (NodeJS express, ReactJS), R (dplyr, tidyverse, ggplot2), Shell
- **Databases:** PostgreSQL (pgvector semantic search, SQL functions and procedures), SQLite